

Acetylcholine activates an inward current in single mammalian smooth muscle cells.

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Acetylcholine, the major excitatory neurotransmitter to the smooth muscle of mammalian intestine, is known to depolarize smooth muscle cells with an apparent increase in membrane conductance. However, the ionic mechanisms that are triggered by muscarinic receptor activation and underlie this response are poorly understood, due in part to the technical problems associated with the electrophysiological study of smooth muscle. The muscarinic action of acetylcholine in certain neurones has been shown to involve the switching off of a resting K⁺ current (M-current) and a similar mechanism has recently also been identified in smooth muscle of amphibian stomach. We have now applied the patch-clamp technique to single smooth muscle cells of rabbit jejunum and find that muscarinic receptor activation switches on a nonselective, voltage-sensitive inward current. In addition, acetylcholine activates and then suppresses spontaneous K⁺ current transients, which are probably triggered by rises in intracellular Ca²⁺ in these cells.

Motoneuron-specific loss of VACHT mimics neuromuscular defects seen in congenital myasthenic syndrome.

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Motoneurons (MNs) control muscle activity by releasing the neurotransmitter acetylcholine (ACh) at the level of neuromuscular junctions. ACh is packaged into synaptic vesicles by the vesicular ACh transporter (VACHT), and disruptions in its release can impair muscle contraction, as seen in congenital myasthenic syndromes (CMS). Recently, VACHT gene mutations were identified in humans displaying varying degrees of myasthenia. Moreover, mice with a global deficiency in VACHT expression display several characteristics of CMS. Despite these findings, little is known about how a long-term decrease in VACHT expression in vivo affects MNs structure and function. Using Cre-loxP technology, we generated a mouse model where VACHT is deleted in select groups of MNs (mnVACHT-KD). Molecular analysis revealed that the VACHT deletion was specific to MNs and affected approximately 50% of its population in the brainstem and spinal cord, with alpha-MNs primarily targeted (70% in spinal cord). Within each animal, the cell body area of VACHT-deleted MNs was significantly smaller compared to MNs with VACHT preserved. Likewise, muscles innervated by VACHT-deleted MNs showed atrophy while muscles innervated by VACHT-containing neurons appeared normal. In addition, mnVACHT KD mice had decreased muscle strength, were hypoactive, leaner and exhibited kyphosis. This neuromuscular dysfunction was evident at 2 months of age and became progressively worse by 6 months. Treatment of mutants with a cholinesterase inhibitor was able to improve some of the motor deficits. As these observations mimic what is seen in CMS, this new line could be valuable for assessing the efficacy of potential CMS drugs.

Purinergic Tuning of the Tripartite Neuromuscular Synapse.

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The vertebrate neuromuscular junction (NMJ) is a specialised chemical synapse involved in the transmission of bioelectric signals between a motor neuron and a skeletal muscle fiber, leading to muscle contraction. Typically, the NMJ is a tripartite synapse comprising (a) a presynaptic region represented by

the motor nerve ending, (b) a postsynaptic skeletal motor endplate area, and (c) perisynaptic Schwann cells (PSCs) that shield the motor nerve terminal. Increasing evidence points towards the role of PSCs in the maintenance and control of neuromuscular integrity, transmission, and plasticity. Acetylcholine (ACh) is the main neurotransmitter at the vertebrate skeletal NMJ, and its role is fine-tuned by co-released purinergic neuromodulators, like adenosine 5'-triphosphate (ATP) and its metabolite adenosine (ADO). Adenine nucleotides modulate transmitter release and expression of postsynaptic ACh receptors at motor synapses via the activation of P2Y and P2X receptors. Endogenously generated ADO modulates ACh release by acting via co-localised inhibitory A1 and facilitatory A2A receptors on motor nerve terminals, whose tonic activation depends on the neuronal firing pattern and their interplay with cholinergic receptors and neuropeptides. Thus, the concerted action of adenine nucleotides, ADO, and ACh/neuropeptide co-transmitters is paramount to adapting the neuromuscular transmission to the working load under pathological conditions, like Myasthenia gravis. Unravelling these functional complexities prompted us to review our knowledge about the way purines orchestrate neuromuscular transmission and plasticity in light of the tripartite synapse concept, emphasising the often-forgotten role of PSCs in this context.

Principles of neuromuscular transmission.

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Activation of skeletal muscles, small and large alike, occurs when spinal cord neurons send action potentials to groups of fibers. The central role of acetylcholine and its receptors in this activation is well established. The mechanisms involved and the implications for understanding neuromuscular disorders and for rationalizing and improving therapy are discussed.

Three-dimensional neuron-muscle constructs with neuromuscular junctions.

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This paper describes a fabrication method of muscle tissue constructs driven by neurotransmitters released from activated motor neurons. The constructs consist of three-dimensional (3D) free-standing skeletal muscle fibers co-cultured with motor neurons. We differentiated mouse neural stem cells (mNSCs) cultured on the skeletal muscle fibers into neurons that extend their processes into the muscle fibers. We found that acetylcholine receptors (AChRs) were formed at the connection between the muscle fibers and the neurons. The neuron-muscle constructs consist of highly aligned, long and matured muscle fibers that facilitate wide contractions of muscle fibers in a single direction. The contractions of the neuron-muscle construct were observed after glutamic acid activation of the neurons. The contraction was stopped by treatment with curare, an neuromuscular junction (NMJ) antagonist. These results indicate that our method succeeded in the formation of NMJs in the neuron-muscle constructs. The neuron-muscle construct system can potentially be used in pharmacokinetic assays related to NMJ disease therapies and in soft-robotic actuators.

Molecular mechanisms of cholinergic neurotransmission in visceral smooth muscles

with a focus on receptor-operated TRPC4 channel and impairment of gastrointestinal motility by general anaesthetics and anxiolytics.

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Acetylcholine is the primary excitatory neurotransmitter in visceral smooth muscles, wherein it binds to and activates two muscarinic receptors subtypes, M2 and M3, thus causing smooth muscle excitation and contraction. The first part of this review focuses on the types of cells involved in cholinergic neurotransmission and on the molecular mechanisms underlying acetylcholine-induced membrane depolarisation, which is the central event of excitation-contraction coupling causing Ca²⁺ entry via L-type Ca²⁺ channels and smooth muscle contraction. Studies of the muscarinic cation current in intestinal myocytes (mICAT) revealed its main molecular counterpart, receptor-operated TRPC4 channel, which is activated in synergy by both M2 and M3 receptors. M3 receptors activation is of permissive nature, while activation of M2 receptors via Gi/o proteins that are coupled to them plays a direct role in TRPC4 opening. Our understanding of signalling pathways underlying mICAT generation has vastly expanded in recent years through studies of TRPC4 gating in native cells and its regulation in heterologous cells. Recent studies using muscarinic receptor knockout have established that at low agonist concentration activation of both M2 receptor and the M2/M3 receptor complex elicits smooth muscle contraction, while at high agonist concentration M3 receptor function becomes dominant. Based on this knowledge, in the second part of this review we discuss the cellular and molecular mechanisms underlying the numerous anticholinergic effects on neuroactive drugs, in particular general anaesthetics and anxiolytics, which can significantly impair gastrointestinal motility. This article is part of the Special Issue on "Ukrainian Neuroscience".

Clinical neurophysiology of neuromuscular junction disease.

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The neuromuscular junction (NMJ) is a cholinergic synapse where quantal release of acetylcholine (ACh) from motor nerve terminals generates a local endplate potential (EPP) on the muscle fiber. EPPs that reach threshold depolarize the entire muscle fiber and initiate the process of excitation-contraction coupling. Deficits of neuromuscular transmission result in clinical weakness that is fatigable and may fluctuate. Repetitive nerve stimulation (RNS) testing can unmask the reduced safety factor common to all NMJ disorders via depletion of immediate ACh stores at the presynaptic motor nerve terminal with decremental responses to low-frequency RNS (LF-RNS). The facilitated responses characterizing presynaptic NMJ disorders can be revealed by brief exercise or high stimulation rates that augment presynaptic calcium levels. Activation with isometric exercise may increase the sensitivity of RNS testing. Attention to technical detail and reproducibility of findings are essential in generating valid results in RNS testing. Motor unit potential (MUP) instability or jiggle is the main finding seen in NMJ disorders on conventional needle EMG and reflects the moment-to-moment variability in the number and synchrony of muscle fiber action potentials (MFAPs) that compose a MUP. Single fiber EMG (SFEMG) is a highly selective technique that assesses jitter, the temporal variability in MFAPs generated in response to motor nerve action potentials.

Simplified in vitro engineering of neuromuscular junctions between rat embryonic

motoneurons and immortalized human skeletal muscle cells.

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Neuromuscular junctions (NMJs) consist of the presynaptic cholinergic motoneuron terminals and the corresponding postsynaptic motor endplates on skeletal muscle fibers. At the NMJ the action potential of the neuron leads, via release of acetylcholine, to muscle membrane depolarization that in turn is translated into muscle contraction and physical movement. Despite the fact that substantial NMJ research has been performed, the potential of in vivo NMJ investigations is inadequate and difficult to employ. A simple and reproducible in vitro NMJ model may provide a robust means to study the impact of neurotrophic factors, growth factors, and hormones on NMJ formation, structure, and function. This report characterizes a novel in vitro NMJ model utilizing immortalized human skeletal muscle stem cells seeded on 35 mm glass-bottom dishes, cocultured and innervated with spinal cord explants from rat embryos at ED 13.5. The cocultures were fixed and stained on day 14 for analysis and assessment of NMJ formation and development. This unique serum- and trophic factor-free system permits the growth of cholinergic motoneurons, the formation of mature NMJs, and the development of highly differentiated contractile myotubes, which exhibit appropriate configuration of transversal triads, representative of in vivo conditions. This coculture system provides a tool to study vital features of NMJ formation, regulation, maintenance, and repair, as well as a model platform to explore neuromuscular diseases and disorders affecting NMJs.

The Neuromuscular Junction in Health and Disease: Molecular Mechanisms Governing Synaptic Formation and Homeostasis.

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The neuromuscular junction (NMJ) is a highly specialized synapse between a motor neuron nerve terminal and its muscle fiber that are responsible for converting electrical impulses generated by the motor neuron into electrical activity in the muscle fibers. On arrival of the motor nerve action potential, calcium enters the presynaptic terminal, which leads to the release of the neurotransmitter acetylcholine (ACh). ACh crosses the synaptic gap and binds to ACh receptors (AChRs) tightly clustered on the surface of the muscle fiber; this leads to the endplate potential which initiates the muscle action potential that results in muscle contraction. This is a simplified version of the events in neuromuscular transmission that take place within milliseconds, and are dependent on a tiny but highly structured NMJ. Much of this review is devoted to describing in more detail the development, maturation, maintenance and regeneration of the NMJ, but first we describe briefly the most important molecules involved and the conditions that affect their numbers and function. Most important clinically worldwide, are myasthenia gravis (MG), the Lambert-Eaton myasthenic syndrome (LEMS) and congenital myasthenic syndromes (CMS), each of which causes specific molecular defects. In addition, we mention the neurotoxins from bacteria, snakes and many other species that interfere with neuromuscular transmission and cause potentially fatal diseases, but have also provided useful probes for investigating neuromuscular transmission. There are also changes in NMJ structure and function in motor neuron disease, spinal muscle atrophy and sarcopenia that are likely to be secondary but might provide treatment targets. The NMJ is one of the best studied and most disease-prone synapses in the nervous system and it is amenable to in vivo and ex vivo investigation and to systemic therapies that can help restore normal function.

Autoregulation of Acetylcholine Release and Micro-Pharmacodynamic Mechanisms at Neuromuscular Junction: Selective Acetylcholinesterase Inhibitors for Therapy of Myasthenic Syndromes.

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Neuromuscular junctions (NMJs) are directly involved into such indispensable to life processes as respiration and locomotion. However, motor nerve forms only one synaptic contact at each muscle fiber. This unique configuration requires specific properties and constrains to be effective. The very high density of acetylcholine receptors (AChRs) of muscle type in synaptic cleft and an excess of acetylcholine (ACh) released under physiological conditions make this synapse extremely reliable. Nevertheless, under pathological conditions such as myasthenia gravis and congenital myasthenic syndromes, the safety factor can be markedly reduced. Drugs used for short-term symptomatic therapy of these pathological states, cause partial inhibition of cholinesterases (ChEs). These enzymes catalyze the hydrolysis of ACh, thus terminate its action on AChRs. Extension of the lifetime of ACh molecules compensates muscular AChRs abnormalities and, consequently, rescues muscle contractions. In this mini review, we will first outline the functional organization of the NMJ, and then, consider the concept of the safety factor and how it may be changed. This will be followed by a look at autoregulation of ACh release that influences the safety factor of NMJs. Finally, we will consider the morphological features of NMJs as a putative reserve to increase effectiveness of pathological muscle weakness therapy by ChEs inhibitors due to opportunity to use micro-pharmacodynamic mechanisms.
